

Set - 1:-

Q01:-

(a) According to Axiomatic defn of probability:

(i) Probability of event E :-

$$P(E) \geq 0$$

(ii) for 'S' being the sample space of experiment (Normalization) (Set of all possible outcomes)

$$P(S) = 1$$

(iii) If X & Y are mutually exclusive events, (Additivity), then,

$$P(X \cup Y) = P(X) + P(Y)$$

(b) Given, in a bolt factory,

M_1	manufactures	25% of total
M_2	"	35% "
M_3	"	40% "

M_1, M_2 and M_3 have defective percentage as follows: 5%, 4% & 2% respectively.

Probability of defective bolts:-

$$P(D) = P\left(\frac{D}{M_1}\right)P(M_1) + P\left(\frac{D}{M_2}\right)P(M_2) + P\left(\frac{D}{M_3}\right)P(M_3)$$

$$= P\left(\frac{D}{M_1}\right) \times P(M_1) + P\left(\frac{D}{M_2}\right) \times P(M_2) + P\left(\frac{D}{M_3}\right) \times P(M_3)$$

$$= (0.05)(0.25) + (0.04)(0.35) + (0.02)(0.40)$$

$$= 0.0125 + 0.0140 + 0.0080 = 0.0345 \quad \text{Ans.}$$

$$P(M_2/D) = \frac{(0.35) \times 0.04}{0.0345} = 0.406 \quad \text{Ans.}$$

(c)

GivenProbability ~~Distribution~~ ^{Distribution} $f(x) =$

$$f(x) = \begin{cases} cx^2 & ; 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

By law of Total Probability

$$\int_0^1 cx^2 dx = 1$$

$$c \left[\frac{x^3}{3} \right]_0^1 = 1$$

$$c = 3$$

$$P\left(x > \frac{3}{4}\right) = \int_{3/4}^1 cx^2 dx = \int_{3/4}^1 3x^2 dx$$

$$= \left[x^3 \right]_{3/4}^1$$

$$= \left(1 - \frac{27}{64} \right) = \frac{37}{64}$$

$$\therefore P\left(x > \frac{3}{4}\right) = \frac{37}{64}$$

CO2:-(a) Mean & Variance of Binomial Distribution:-

We know,

$$\text{Probability Mass function} = f_x(x) = {}^nC_x p^x q^{n-x}$$

 $p =$ Probability of success

$$p + q = 1$$

 $q =$ failure

Mean = Expectation of X .

$$= \sum_{i=0}^n x_i f_X(x_i)$$

$$= 1 \times {}^nC_1 p^1 q^{n-1} + 2 \times {}^nC_2 p^2 q^{n-2} + \dots + n \times {}^nC_n p^n q^0$$

$$= np [q^{n-1} + {}^{n-1}C_1 + \dots + {}^{n-1}C_{n-1} q^{n-n}]$$

$$= n \times {}^{n-1}C_{n-1} q^{n-1}$$

Bernoulli R.V, y :-

Mean = np .

Ans.

$$y = \begin{cases} 1, & \text{success} \\ 0, & \text{failure} \end{cases}$$

$$\text{Var}(X) = \text{Var}(X_1 + X_2 + \dots + X_n) \quad \text{Exp}(Y) = 1 \times P(Y=1) + 0 \times P(Y=0)$$

$$= \text{Var}(X_1) + \text{Var}(X_2) + \dots + \text{Var}(X_n)$$

$$= 1 \times p + 0 \times q$$

$$= p$$

$\text{Var}(X) = npq$

$$\{X_i = p, q\}$$

$$\text{Var}(Y) = E(Y^2) - (E(Y))^2$$

$$= p(Y^2) - p^2$$

$$\text{Var}(X) = E(X^2) - (E(X))^2$$

$$= E(X(X-1) + X) - (E(X))^2$$

$$= p(1-p)$$

$$= pq$$

$$= n(n-1)p^2 + np - n^2p^2$$

$$= npq$$

(b)

Given, for poisson Distribution,

$$P(X > 0) = 1 - e^{-\lambda} \quad \text{--- (1)}$$

$$P(X > 0) = 1 - P(X = 0) \quad \text{--- (2)}$$

for Poisson Distribution: $P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}$

$$P(X=0) = \frac{e^{-\lambda} \lambda^0}{0!} = e^{-\lambda} \quad \text{--- (3)}$$

from (1), (2) & (3),

$$\lambda = 2$$

$$P(1.5 < X < 1.5) = P(X=0) + P(X=1)$$

$$= e^{-2} + \frac{e^{-2} \cdot 2}{1} = 3e^{-2}$$

Ans.

(a)

Total no. of workers $\rightarrow 120$

Standard Deviation (σ) $\rightarrow 3.03$

Mean (μ) $\rightarrow 11.35$

Normal Distribution

Standardize:

$$Z = \frac{9 - 11.35}{3.03} = -0.78$$

$$Z = \frac{17 - 11.35}{3.03} = 1.86$$

$$P(-0.78 < Z < 1.86)$$

$$= 0.2823 + 0.4688$$

$$= 0.7509$$

% of workers $\rightarrow 75.09\%$

Ans

Q3:-

(a) Given, Lines of Regression:

$$x + 2y = 5 \quad \text{--- (1)}$$

$$2x + 3y = 8 \quad \text{--- (2)}$$

$$\sigma_x^2 = 12$$

$$\bar{y} = 2$$

Means: $\bar{x} = 1$

$$2x + 4y = 10$$

$$\bar{y} = 2, \bar{x} = 1$$

$$\text{①} \Rightarrow y = -\frac{1}{2}x + \frac{5}{2}$$

$$b_{yx} = -\frac{1}{2}$$

Reg coeff of y on x

$$(2) \Rightarrow x = -\frac{3}{2}y + 4$$

$$\hookrightarrow b_{xy} = -\frac{3}{2}$$

$$r^2 = b_{xy} \cdot b_{yx}$$

$$r = \pm \frac{\sqrt{3}}{2}$$

∵ b_{xy} & b_{yx} are -ve;

$$r = -\frac{\sqrt{3}}{2}$$

$$b_{yx} = r \frac{\sigma_y}{\sigma_x}$$

$$-\frac{1}{2} = -\frac{\sqrt{3}}{2} \times \frac{\sigma_y}{\sqrt{3}}$$

$$\sigma_y = 2$$

b) Given, $m_2' \rightarrow$ 2nd moment about origin 'a' μ_x .
 $m_2 \rightarrow$ " " " "

$$\text{To prove: } m_2' = m_2 + d^2$$

$d = \bar{x} - a$

for 2nd Moment:-

$$m_2' = \frac{1}{n} \sum_{i=1}^n (x_i - a)^2$$

$$m_2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

↓

$$m_2' = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x} + \bar{x} - a)^2$$

$$\Rightarrow m_2' = \frac{1}{n} \sum_{i=1}^n ((x_i - \bar{x}) + d)^2$$

$$m_2' = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 + \frac{1}{n} \sum_{i=1}^n d^2 + \frac{2}{n} \sum_{i=1}^n (x_i - \bar{x})d$$

$\downarrow$ this is 0

$$m_2' = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 + \frac{1}{n} \sum_{i=1}^n d^2 + \frac{2d}{n} \sum_{i=1}^n (x_i - \bar{x})$$

$\downarrow$ this is 0

$$m_2' = m_2 + d^2$$

Proved

Set-2:-

Q01:-

- Q1) Two or more events are said to be mutually exhaustive events if ~~at~~ they cover all possible outcomes.

Let $A_1, A_2, \dots, A_n$ be the events.

$$A_1 \cup A_2 \cup \dots \cup A_n = S$$

Ex: Tossing a coin $\rightarrow \{\text{Head, Tail}\}$

$A, B \rightarrow$ independent events, either one is null.

$$[P(A)=0 \text{ or } P(B)=0]$$

$$P(A \cap B) = P(A)P(B)$$

Either Probability $= 0 \rightarrow P(A \cap B) = 0$

Q2) Given,

Bag has 4R & 3B balls.

Two drawings of 2 balls

$\hookrightarrow$ 1st '2' ~~or~~ R, next '2' B.

for 1st Draw of 2R $\Rightarrow \frac{{}^4C_2}{{}^7C_2} = \frac{2}{7}$

Replacement

No replacement

for 2nd Draw of 2B $\Rightarrow \frac{{}^3C_2}{{}^7C_2} = \frac{3}{10}$

$= \frac{1}{7}$

$P_{\text{net}} = \frac{2}{49}$

$\frac{3}{35}$

Ans,

② Given ~~PDF~~ $f(x) = \begin{cases} x & ; 0 \leq x < 1 \\ k-x & ; 1 \leq x \leq 2 \\ 0 & ; \text{o.w.} \end{cases}$

PDF of X is given by

for P.D.F.;

$$\int_0^1 x dx + \int_1^2 (k-x) dx = 1$$

$$\left[\frac{x^2}{2} \right]_0^1 + \left[kx - \frac{x^2}{2} \right]_1^2 = 1$$

$$\frac{1}{2} + (2k-2) - \left(k - \frac{1}{2} \right) = 1$$

$$(2k-1) - (k-1) = 1$$

$$k = 2$$

$$P(X \leq 1.5) =$$

$$\int_0^1 x dx + \int_1^{1.5} (2-x) dx$$

$$= \left[\frac{x^2}{2} \right]_0^1 + \left[2x - \frac{x^2}{2} \right]_1^{1.5}$$

$$= \frac{1}{2} + \left(3 - \frac{9}{8} \right) - \left(2 - \frac{1}{2} \right)$$

$$= \frac{1}{2} + \frac{15}{8} - \frac{3}{2}$$

$$P(X \leq 1.5) = \frac{7}{8}$$

CO2:-

(a) Mean & Variance of Poisson:-

$$X \sim \text{Poisson}(\lambda)$$

$$E(X) = \sum_{x=0}^{\infty} x P(X=x)$$

$$E(X) = \sum_{x=0}^{\infty} x \frac{e^{-\lambda} \lambda^x}{x!}$$

$$E(X) = e^{-\lambda} \sum_{x=0}^{\infty} \frac{x \lambda^x}{x!}$$

$$E(X) = e^{-\lambda} \sum_{x=1}^{\infty} \frac{\lambda^{x-1}}{(x-1)!}$$

$$E(X) = e^{-\lambda} \lambda \sum_{y=0}^{\infty} \frac{\lambda^y}{y!} \quad x-1 \rightarrow y$$

$$E(X) = e^{-\lambda} \lambda e^{\lambda}$$

$$\boxed{E(X) = \lambda}$$

$$\text{Var}(X) = E(X^2) - (E(X))^2$$

$$E(X(X-1)) = \sum x(x-1) P(X=x)$$

$$= e^{-\lambda} \sum \frac{x(x-1) \lambda^x}{x!}$$

$$= \lambda^2 e^{-\lambda} \sum_{x=2}^{\infty} \frac{\lambda^{x-2}}{(x-2)!}$$

$$= \lambda^2 e^{-\lambda} \sum_{y=0}^{\infty} \frac{\lambda^y}{y!}$$

$$= e^{-\lambda} \lambda^2 e^{\lambda}$$

$$\boxed{\text{Var}(X) = \lambda^2}$$

⑥

Overall % = 40.

Out of 6 candidates $\rightarrow$ atleast 4 pass

$$p = 0.6 \quad q = 0.4$$

$$P(X \geq 4)$$

$$= P(4) + P(5) + P(6)$$

$$= {}^6C_4 (0.6)^4 (0.4)^2 + {}^6C_5 (0.6)^5 (0.4)^1 + {}^6C_6 (0.6)^6 (0.4)^0$$

$$P(X \geq 4) = 0.54$$

⑦

Poisson Addition:-

$$\lambda_1 = 2$$

$$\lambda_2 = 6$$

$$\lambda = 2 + 6 = 8$$

Probability that total no. of deaths on particular day is 2 or fewer

$$P(X \leq 2) = P(X=0) + P(X=1) + P(X=2)$$

$$= \frac{e^{-\lambda} \lambda^0}{0!} + \frac{e^{-\lambda} \lambda^1}{1!} + \frac{e^{-\lambda} \lambda^2}{2!}$$

$$= e^{-8} \left[\frac{8^0}{0!} + \frac{8^1}{1!} + \frac{8^2}{2!} \right]$$

$$= e^{-8} (1 + 8 + 32)$$

$$P(X \leq 2) = e^{-8} 41$$

(3) a

$$x + 4y + 3 = 0 \quad (1)$$

$$4x + 9y + 5 = 0 \quad (2)$$

$$\text{by } x = -\frac{4}{9}$$

$$\text{by } y = -\frac{4}{9}$$

$$(1) \times 4$$

$$4x + 16y + 12 = 0$$

$$4x + 9y + 5 = 0$$

$$\begin{array}{r} \xrightarrow{+} \quad \xrightarrow{-} \quad \xrightarrow{-} \end{array}$$

$$\boxed{\bar{x} = -7}$$

$$\boxed{\bar{y} = +1}$$

$$r^2 = \frac{16}{9}$$

$$r = \pm \frac{4}{3} > 1$$

$$\text{(II) by } x = -\frac{1}{4}$$

$$\text{by } y = -\frac{9}{4}$$

$$r^2 = \frac{9}{16}$$

$$r = \pm \frac{3}{4}$$

$$\therefore r = -0.75$$

$$[\because \text{by } x, \text{ by } y \neq 0]$$

(b)

$$A = 7$$

$$\mu_1' = 0.2, \mu_2' = 19.4, \mu_3' = -11.0$$

$$\bar{x} = A + \bar{\mu}_1$$

$$\boxed{\bar{x} = 7.2}$$

$$S.D = \sigma = \sqrt{\mu_2}$$

$$= 4.4$$

$$\mu_2 = \mu_2' - (\mu_1')^2$$

$$\mu_2 = 19.4 - (0.2)^2$$

$$\boxed{\mu_2 = 19.36}$$

↓ variance

Q01:-

- (a) ~~two~~ when two events E_1, E_2 occur simultaneously they are said to be compound events & their probability is called compound probability.

Ex: two dice are thrown simultaneously.

- (b) Urn I: 1W, 2B, 3R.
 Urn II: 2W, 1B, 1R.
 Urn III: 4W, 5B, 3R.

One urn chosen $\rightarrow$ two balls are drawn (w & R)

$$P(U_1) = P(U_2) = P(U_3) = \frac{1}{3}$$

$$P(W, R) = P(W \text{ then } R) + P(R \text{ then } W)$$

$$P(W, R | U_1) = \frac{1}{6} \cdot \frac{3}{5} + \frac{3}{6} \cdot \frac{1}{5} = \frac{1}{5}$$

$$P(W, R | U_2) = \frac{2}{4} \cdot \frac{1}{3} + \frac{1}{4} \cdot \frac{2}{3} = \frac{1}{3}$$

$$P(W, R | U_3) = \frac{4}{12} \cdot \frac{3}{11} + \frac{3}{12} \cdot \frac{4}{11} = \frac{2}{11}$$

$$P(W, R) = \frac{1}{3} \left(\frac{1}{5} + \frac{1}{3} + \frac{2}{11} \right) = \frac{118}{495}$$

$$P(U_2 | W, R) = \frac{P(U_2) \cdot P(W, R | U_2)}{P(W, R)}$$

$$= \frac{\frac{1}{3} \times \frac{1}{3}}{\frac{118}{495}} = \frac{55}{118}$$

$$\boxed{P(U_2 | W, R) = \frac{55}{118}}$$

(c)

$$f(x) = ax; \quad x = 1, 2, 3, \dots, n$$

p.m.f of

Discrete R.V.X

for p.m.f, ~~(a) (b) (c) (d)~~

$$\sum_{x=1}^n ax = 1.$$

$$a \frac{n(n+1)}{2} = 1.$$

$$a = \frac{2}{n(n+1)}$$

$$E(X) = \sum_{x=1}^n x f(x)$$

$$= \sum_{x=1}^n ax^2$$

$$= a \frac{n(n+1)(2n+1)}{6}$$

$$E(X) = \frac{2n+1}{3}$$

$$[E(X^2)] = \sum_{x=1}^n x^2 f(x)$$

$$= a \sum_{x=1}^n x^3$$

$$= \frac{2}{n(n+1)} \left(\frac{n(n+1)}{2} \right)^2$$

$$= \frac{n(n+1)}{2}$$

$$\text{Var}(X) = E(X^2) - (E(X))^2$$

$$\text{Var}(X) = \frac{n(n+1)}{2} - \left(\frac{2n+1}{3} \right)^2$$

(a) (i)

Binomial:

$$P(X=x) = {}^n C_x p^x q^{n-x}$$

$$\therefore \lambda = pn \Rightarrow p = \frac{\lambda}{n}$$

$$q = 1 - p = 1 - \frac{\lambda}{n}$$

$$P(X=x) = {}^n C_x \left(\frac{\lambda}{n}\right)^x \left(1 - \frac{\lambda}{n}\right)^{n-x}$$

$$P(X=x) = \frac{n(n-1)(n-2)\dots(n-x+1)}{x!} \frac{\lambda^x}{n^x} \left(1 - \frac{\lambda}{n}\right)^{n-x}$$

$$P(X=x) = \frac{n}{x} \cdot \frac{n-1}{n} \cdot \dots \cdot \frac{n-x+1}{n} \frac{\lambda^x}{x!} \left(1 - \frac{\lambda}{n}\right)^{n-x}$$

$$P(X=x) = \frac{\lambda^x}{x!} \left(1 - \frac{\lambda}{n}\right)^n \left(1 - \frac{\lambda}{n}\right)^{-x}$$

$$\boxed{P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}} \quad (\text{Poisson})$$

(ii) Poisson $\longrightarrow$ Normal.
 $X \sim \text{Poisson}(\lambda)$

Mean = Variance = λ

$$Z = \frac{X - \lambda}{\sqrt{\lambda}} \quad \text{S.D.} = \sqrt{\lambda}$$

$$P(X \leq x) \longrightarrow P\left(Z \leq \frac{x + 0.5 - \lambda}{\sqrt{\lambda}}\right)$$

$$\text{Poisson}(\lambda) \approx N(\lambda, \lambda)$$

(b)

Given,
 $P(X=1) = P(X=2)$
 $\frac{e^{-\lambda} \lambda^1}{1!} = \frac{e^{-\lambda} \lambda^2}{2!}$

$\lambda = 2$

Mean = Variance = 2

(c)

C, Java, Python.

Mean: 50, 52, 48.

S.D: 15, 12, 16.

$Z = \frac{x - \mu}{\sigma}$

Total: $\mu = 150$

$\sigma^2 = 15^2 + 12^2 + 16^2 = 625$

$\sigma = 25$

for 90:

$Z = \frac{90 - 150}{25} = -2.4$

for 180:

$Z = \frac{180 - 150}{25} = 1.2$

$P(-2.4 < Z < 1.2) = 0.8849 - (1 - 0.9918)$
 $= 0.8767$

Ans:-

(a)

$x = -0.2y + 4.2$

$y = -0.8x + 8.4$

$r^2 = 0.16$

$r = \pm 0.4 \rightarrow -0.4$

$(\because b_{xy}, b_{yx} < 0)$

$$y = -0.8(4) + 8.4 = 5.2$$

⑥ Moments about 4:-

$$E(x-4) = -1.5$$

$$E[(x-4)^2] = 2.7$$

$$E(x) - 4 = -1.5$$

$$\boxed{\mu = E(x) = 2.5}$$

$$E[(x-4)^2] = \text{Var}(x) + (\mu-4)^2$$

$$2.7 = \sigma^2 + (2.5-4)^2$$

$$\sigma^2 = 0.45$$

$$\boxed{\sigma = 0.67}$$

← x →